

AKTU B.Tech Computer Science & Engineering - 3rd Semester

1. Semester 3 Evaluation Scheme & Credits

- **Universal Human Value and Professional Ethics (BVE301):** 3 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Data Structure (BCS301):** 4 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Computer Organization and Architecture (BCS302):** 4 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Discrete Structures & Theory of Logic (BCS303):** 3 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Science Based Open Elective (BOE3):**** 4 Credits | Theory (100 or 70 Marks).
 - **Data Structure Lab (BCS351):** 1 Credit | Practical (50 Marks Internal, 50 Marks External) | Total: 100 Marks.
 - **Computer Organization and Architecture Lab (BCS352):** 1 Credit | Practical (50 Marks) | Total: 50 Marks.
 - **Web Designing Workshop (BCS353):** 1 Credit | Practical (50 Marks) | Total: 50 Marks.
 - **Cyber Security (BCC301):** 2 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Total Semester Marks:** 900 Marks.
 - **Total Semester Credits:** 25 Credits.
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2. Detailed Syllabus - Core Theory Subjects

2.1 Data Structure (BCS301)

- **Subject Code:** BCS301
- **Credits:** 4 Credits
- **Unit 1: Introduction to Arrays and Linked Lists**
 - Topics: Basic Terminology, Elementary Data Organization, Built in Data Types in C.
 - Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Big Oh, Big Theta and Big Omega, Time-Space trade-off.
 - Abstract Data Types (ADT), Arrays: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D, 2-D, 3-D and n-D Array, Application of arrays, Sparse Matrices and their representations.

- Linked lists: Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List: Insertion, Deletion, Traversal, Polynomial Representation and Addition, Subtraction & Multiplications of Single variable & Two variables Polynomial.
 - **Unit 2: Stacks and Queues**
 - Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression.
 - Recursion: Iteration and Recursion- Principles of recursion, Tail recursion, Removal of recursion, Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Tradeoffs between iteration and recursion.
 - Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.
 - **Unit 3: Searching and Sorting**
 - Searching: Concept of Searching, Sequential search, Index Sequential Search, Binary Search.
 - Hashing: Concept of Hashing & Collision resolution Techniques used in Hashing.
 - Sorting: Insertion Sort, Selection, Bubble Sort, Quick Sort, Merge Sort, Heap Sort and Radix Sort.
 - **Unit 4: Trees**
 - Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Strictly Binary Tree, Complete Binary Tree, Extended Binary Trees.
 - Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search Tree.
 - Advanced Trees: Threaded Binary trees, Traversing Threaded Binary trees. Huffman coding using Binary Tree. Concept & Basic Operations for AVL Tree, B Tree & Binary Heaps.
 - **Unit 5: Graphs**
 - Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency.
 - Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm.
 - Shortest Path: Transitive Closure and Shortest Path algorithm: Warshal Algorithm and Dijkstra Algorithm.
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2.2 Computer Organization and Architecture (BCS302)

- **Subject Code:** BCS302
 - **Credits:** 4 Credits
 - **Unit 1: Introduction to Digital Computer System**
 - Topics: Functional units of digital system and their interconnections, buses, bus architecture, types of buses and bus arbitration.
 - Register, bus and memory transfer.
 - Processor organization, general registers organization, stack organization and addressing modes.
 - **Unit 2: Arithmetic and Logic Unit (ALU)**
 - Topics: Look ahead carry adders.
 - Multiplication: Signed operand multiplication, Booth's algorithm and array multiplier.
 - Division and logic operations.
 - Floating point arithmetic operation, Arithmetic & logic unit design, IEEE Standard for Floating Point Numbers.
 - **Unit 3: Control Unit and Pipelining**
 - Topics: Instruction types, formats, instruction cycles and sub cycles (fetch and execute etc), micro operations, execution of a complete instruction.
 - Program Control, Reduced Instruction Set Computer (RISC), Pipelining.
 - Hardwired and microprogrammed control: microprogramme sequencing, concept of horizontal and vertical microprogramming.
 - **Unit 4: Memory Architecture**
 - Topics: Basic concept and hierarchy, semiconductor RAM memories, 2D & 2 1/2D memory organization, ROM memories.
 - Cache memories: concept and design issues & performance, address mapping and replacement.
 - Auxiliary memories: magnetic disk, magnetic tape and optical disks. Virtual memory: concept implementation.
 - **Unit 5: Input/Output Organization**
 - Topics: Peripheral devices, I/O interface, I/O ports.
 - Interrupts: interrupt hardware, types of interrupts and exceptions.
 - Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access (DMA), I/O channels and processors.
 - Serial Communication: Synchronous & asynchronous communication, standard communication interfaces.
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2.3 Discrete Structures & Theory of Logic (BCS303)

- **Subject Code:** BCS303
- **Credits:** 3 Credits
- **Unit 1: Set Theory, Relations, POSET & Lattices**
 - Set Theory & Relations: Introduction, Combination of sets. Relations: Definition, Operations on relations, Properties of relations, Composite Relations, Equality of relations, Recursive definition of relation, Order of relations.

- POSET & Lattices: Hasse Diagram, POSET, Definition & Properties of lattices - Bounded, Complemented, Distributed, Modular and Complete lattice.
- **Unit 2: Functions & Boolean Algebra**
 - Functions: Definition, Classification of functions, Operations on functions. Growth of Functions.
 - Boolean Algebra: Introduction, Axioms and Theorems of Boolean algebra, Algebraic manipulation of Boolean expressions. Simplification of Boolean Functions, Karnaugh maps.
- **Unit 3: Theory of Logics**
 - Topics: Proposition, Truth tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Theory of Inference.
 - Predicate Logic: First order predicate, well-formed formula of predicate, quantifiers, Inference theory of predicate logic.
- **Unit 4: Algebraic Structures**
 - Topics: Definition, Groups, Subgroups and order, Cyclic Groups, Cosets, Lagrange's theorem, Normal Subgroups, Permutation and Symmetric groups, Group Homomorphisms, Definition and elementary properties of Rings and Fields.
- **Unit 5: Graphs & Combinatorics**
 - Graphs: Definition and terminology, Representation of graphs, Multigraphs, Bipartite graphs, Planar graphs, Isomorphism and Homeomorphism of graphs, Euler and Hamiltonian paths, Graph coloring.
 - Combinatorics: Introduction, Counting Techniques, Pigeonhole Principle.

AKTU B.Tech Computer Science & Engineering - 4th Semester

1. Semester 4 Evaluation Scheme & Credits

- **Technical Communication (BAS401):** 3 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
- **Operating System (BCS401):** 4 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
- **Theory of Automata and Formal Languages (BCS402):** 4 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
- **Object Oriented Programming with Java (BCS403):** 3 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
- **BSC Maths-IV (BAS403):** 4 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
- **Operating System Lab (BCS451):** 1 Credit | Practical (50 Marks Internal, 50 Marks External) | Total: 100 Marks.
- **Object Oriented Programming with Java Lab (BCS452):** 1 Credit | Practical (50 Marks Internal, 50 Marks External) | Total: 100 Marks.

- **Cyber Security Workshop (BCS453):** 1 Credit | Practical (50 Marks Internal, 50 Marks External) | Total: 100 Marks.
 - **Python Programming (BCC402/):** 2 Credits | Theory (70 Marks), Internals (30 Marks) | Total: 100 Marks.
 - **Sports and Yoga-II / NSS-II (BVE451/BVE452):** 0 Credits (Audit Course) | Practical (100 Marks).
 - **Total Semester Marks:** 800 Marks.
 - **Total Semester Credits:** 23 Credits.
 - **Important Note:** Mini Project ya 4-weeks ki internship summer break (4th semester ke baad) ke dauran karni mandatory hai, jiska assessment 5th semester mein hota hai.
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2. Detailed Syllabus - Core Theory Subjects

2.1 Operating System (BCS401)

- **Subject Code:** BCS401
- **Credits:** 4 Credits
- **Unit 1: Introduction to OS & Structures**
 - Topics: Operating system and functions, Classification of Operating systems - Batch, Interactive, Time sharing, Real Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems.
 - Operating System Structure: Layered structure, System Components, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.
- **Unit 2: Concurrent Processes**
 - Topics: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation.
 - Classical Problems in Concurrency: Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models and Schemes, Process generation.
- **Unit 3: CPU Scheduling & Deadlock**
 - CPU Scheduling: Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Threads and their management, Scheduling Algorithms, Multiprocessor Scheduling.
 - Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock.
- **Unit 4: Memory Management**
 - Topics: Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation.
 - Virtual Memory: Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.

- **Unit 5: I/O Management & File Systems**
 - I/O Management: I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling, RAID.
 - File System: File concept, File organization and access mechanism, File directories, and File sharing, File system implementation issues, File system protection and security.
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2.2 Theory of Automata and Formal Languages (BCS402)

- **Subject Code:** BCS402
- **Credits:** 4 Credits
- **Unit 1: Basic Concepts and Automata Theory**
 - Topics: Introduction to Theory of Computation - Automata, Computability and Complexity, Alphabet, Symbol, String, Formal Languages.
 - Deterministic Finite Automaton (DFA): Definition, Representation, Acceptability of a String and Language.
 - Non-Deterministic Finite Automaton (NFA): Definition, Equivalence of DFA and NFA, NFA with ϵ -Transition, Equivalence of NFA's with and without ϵ -Transition.
 - Finite Automata with Output: Moore Machine, Mealy Machine, Equivalence of Moore and Mealy Machine, Minimization of Finite Automata.
- **Unit 2: Regular Expressions and Languages**
 - Topics: Regular Expressions, Transition Graph, Kleene's Theorem, Finite Automata and Regular Expression - Arden's theorem, Algebraic Method Using Arden's Theorem.
 - Regular and Non-Regular Languages: Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability - Decision properties, Finite Automata and Regular Languages.
- **Unit 3: Regular and Non-Regular Grammars**
 - Topics: Context Free Grammar (CFG) - Definition, Derivations, Languages, Derivation Trees and Ambiguity.
 - Regular Grammars: Right Linear and Left Linear grammars, Conversion of FA into CFG and Regular grammar into FA, Simplification of CFG.
 - Normal Forms: Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky Hierarchy, Programming problems based on the properties of CFGs.
- **Unit 4: Push Down Automata & Context Free Languages**
 - Pushdown Automata: Nondeterministic Pushdown Automata (NPDA) - Definition, Moves, A Language Accepted by NPDA, Deterministic Pushdown Automata (DPDA) and Deterministic Context free Languages (DCFL), Pushdown Automata for Context Free Languages, Context Free grammars for Pushdown Automata, Two stack Pushdown Automata.
 - Properties of CFL: Pumping Lemma for CFL, Closure properties of CFL, Decision Problems of CFL, Programming problems based on the properties of CFLs.
- **Unit 5: Turing Machines and Recursive Function Theory**

- Turing Machines: Basic Turing Machine Model, Representation of Turing Machines, Language Acceptability of Turing Machines, Techniques for Turing Machine Construction, Modifications of Turing Machine, Turing Machine as Computer of Integer Functions, Universal Turing machine, Linear Bounded Automata.
 - Theory: Church's Thesis, Recursive and Recursively Enumerable language, Halting Problem, Post's Correspondence Problem, Introduction to Recursive Function Theory.
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2.3 Object Oriented Programming with Java (BCS403)

- **Subject Code:** BCS403
- **Credits:** 3 Credits
- **Unit 1: Java Basics & Structures**
 - Fundamentals: Why Java, History of Java, JVM, JRE, Java Environment, Java Source File Structure, and Compilation.
 - Programming Structures: Defining Classes in Java, Constructors, Methods, Access Specifiers, Static Members, Final Members, Comments, Data types, Variables, Operators, Control Flow, Arrays & String.
- **Unit 2: OOP Principles, Packages & Exception Handling**
 - OOP Concepts: Class, Object, Inheritance, Super Class, Sub Class, Overriding, Overloading, Encapsulation, Polymorphism, Abstraction, Interfaces, and Abstract Class.
 - Packages: Defining Package, CLASSPATH Setting for Packages, Making JAR Files for Library Packages, Import and Static Import Naming Convention For Packages.
 - Exception Handling: The Idea behind Exception, Exceptions & Errors, Types of Exception, Control Flow in Exceptions, JVM Reaction to Exceptions, Use of try, catch, finally, throw, throws in Exception Handling, In-built and User Defined Exceptions, Checked and Un-Checked Exceptions.
- **Unit 3: I/O Basics, Multithreading & Java New Features**
 - Input/Output Basics: Byte Streams and Character Streams, Reading and Writing File in Java.
 - Multithreading: Thread, Thread Life Cycle, Creating Threads, Thread Priorities, Synchronizing Threads, Inter-thread Communication.
 - Java New Features: Functional Interfaces, Lambda Expression, Method References, Stream API, Default Methods, Static Method, Base64 Encode and Decode, ForEach Method, Try-with-resources, Type Annotations, Repeating Annotations, Java Module System, Diamond Syntax with Inner Anonymous Class, Local Variable Type Inference, Switch Expressions, Yield Keyword, Text Blocks, Records, Sealed Classes.
- **Unit 4: Java Collections Framework**
 - Topics: Collection in Java, Collection Framework in Java, Hierarchy of Collection Framework, Iterator Interface, Collection Interface, List Interface, ArrayList, LinkedList, Vector, Stack, Queue Interface, Set Interface, HashSet, LinkedHashSet,

SortedSet Interface, TreeSet, Map Interface, HashMap Class, LinkedHashMap Class, TreeMap Class, Hashtable Class, Sorting, Comparable Interface, Comparator Interface, Properties Class in Java.

- **Unit 5: Spring Framework & Spring Boot**

- Spring Core Basics: Spring Dependency Injection concepts, Spring Inversion of Control (IoC), AOP, Bean Scopes - Singleton, Prototype, Request, Session, Application, Web Socket, Auto wiring, Annotations, Life Cycle Call backs, Bean Configuration styles.
- Spring Boot: Spring Boot Build Systems, Spring Boot Code Structure, Spring Boot Runners, Logger.
- Web Services: BUILDING RESTFUL WEB SERVICES, Rest Controller, Request Mapping, Request Body, Path Variable, Request Parameter, GET, POST, PUT, DELETE APIs, Build Web Applications.